

Chap. 49 Modes

3. Modes on Regular Triangular Drum

We have learnt in the Lecture that the circular frequency ω of vibration mode on a square drum of the side length a is given by

$$\omega^2 = \left(\frac{\pi c}{a}\right)^2 (m^2 + n^2),$$

where c is the sound velocity in the drumhead, and m, n are positive integers labeling the modes.

(a) Find the density of states (the number of modes per unit circular frequency per unit area) of the square drum around a given ω , in the continuum limit (i.e. $\omega \gg \pi c/a$).

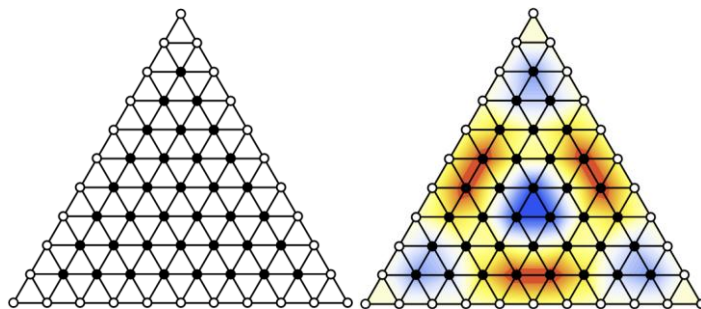
Now let us consider a regular triangular drum with side length a . The mode pattern of the displacement ϕ of the triangular drumhead is determined by the eigen problem

$$-c^2 \nabla^2 \phi = \omega^2 \phi,$$

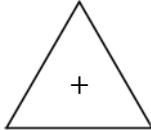
subject to the boundary condition that ϕ must vanish on the triangular edge.

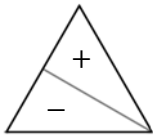
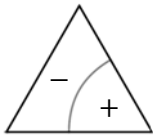
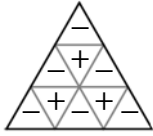
(b) Design a computer program to solve the eigen problem. Draw the patterns of the first 11 modes and find their corresponding frequencies.

Hint: You may discretize the drumhead into a grid of little triangles as shown below. Label the grid points, and collect the displacement field ϕ on them into a column vector.



Then discretize the operator ∇^2 on the triangular lattice to its matrix representation, such that it can be applied to the vector of displacement field, just as applying ∇^2 operator to ϕ . Thus the eigen problem of a partial differential equation is reduced to the diagonalization of the matrix representation of ∇^2 , which can be easily solved by a mathematical software in few seconds. The boundary points (hollow circles in the figure) should be treated specially to ensure the open boundary condition. The trick is to assign very large diagonal elements for the boundary points, such that the boundary states will be raised to a subspace of extremely high momentum, hence being effectively projected out in the low momentum subspace which we concern. You are expected to obtain the following result:

Mode shape	m	n	$(\omega/\omega_0)^2$	ω/ω_0
	1	1	5.33	2.31

	1	2	12.44	3.53
	2	1	12.44	3.53
...?	...?	...?	...?	...?
	3	3	48.00	6.93

Here the circular frequency is measured in the unit $\omega_0 \equiv \pi c/a$.

(c) Use your numerical result to check that the mode frequencies of the triangular drum are given by the following formula:

$$\omega^2 = \left(\frac{4\pi c}{3a}\right)^2 (m^2 + mn + n^2),$$

with m and n both positive integers.

Hint: For those who wish to deduce this formula analytically, just generalize Feynman's plane wave reflection arguments in the Lecture. Note that on the triangular drum, we have six waves to be added together instead of four for the square drum.

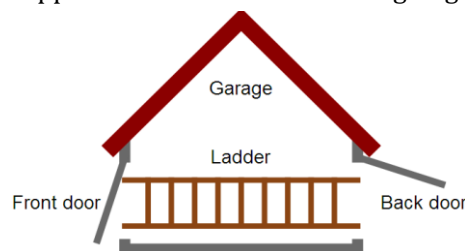
(d) Find the density of states of the triangular drum in the continuum limit. Compare it with the square drum case.

Although the frequency spectrums are totally different for different shape of drums, their densities of states somehow remain the same. This is an important property for the density of state. It has a notable consequence in thermodynamic, that the thermal behaviors of gas (e.g. the ideal gas law $pV = nRT$) are totally independent of the shape of the container. No matter square or round, only the volume matters.

Chap. 15 The Special Theory of Relativity

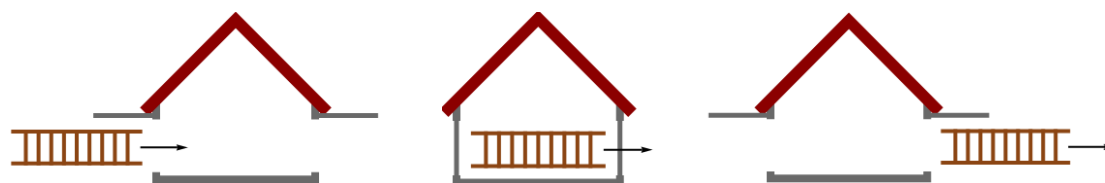
1. Ladder Paradox

Suppose we have a ladder and a garage of the same length L , when both of them are at rest.



Now consider the ladder is traveling at a speed of $0.6c$. From the frame of reference of the garage, its length is contracted to less than the length of the garage. Then from the frame of reference of the garage, there is a moment in time when the ladder can fit completely inside

the garage, and during that moment one can close and open both front and back doors of the garage, without affecting the ladder.

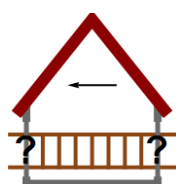


Ladder enters from the front door.

Both doors shut simultaneously.

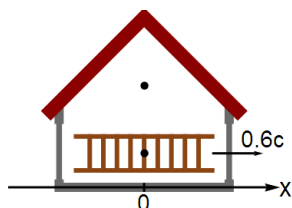
Both doors soon open again and let the ladder out.

However, from the frame of reference of the ladder, now it is the garage that is shorter than the length of the ladder. Therefore there is no moment in time when the ladder is completely inside the garage; and there is no moment when one can close both doors without it hitting the ladder. This is an apparent paradox.



(a) To solve this paradox, first work out what is the length of the ladder from the frame of the garage, and what is the length of the garage from the frame of the ladder.

(b) From the garage frame, let $t = 0$ be the moment when two doors close and open. At this moment, the ladder is right at the middle of the garage which is set to be the origin of coordinate x .



Mark the following figures on the space-time (t v.s. x) diagram from the garage frame:

- (1) the world line of the front door,
- (2) the world line of the back door,
- (3) the world line of the left end of the ladder,
- (4) the world line of the right end of the ladder,
- (5) the event A: front door closes and opens,
- (6) the event B: back door closes and opens,
- (7) the event C: the left end of the ladder passes by the front door,
- (8) the event D: the right end of the ladder passes by the back door.

(c) For the above events A, B, C and D, write down their space-time coordinates (in the form of (x, t)) from the garage frame of reference.

(d) Use Lorentz transformation to transform the space-time coordinates events A, B, C and D into the ladder frame of reference.

The solution to the ladder paradox lies in the fact that what one observer (e.g. the garage) considers as simultaneous does not correspond to what the other observer (e.g. the ladder) considers as simultaneous (relative simultaneity). You will find that events A and B are

simultaneous from the garage frame, but not simultaneous from the ladder frame. Between A and B, which event happens earlier from the ladder frame?

(e) Now mark the world lines and events listed in problem (b) on the space-time diagram from the ladder frame.

(f) What is the scenario from the ladder's point of view?

(g)* What if the back door is closed permanently and does not open again? Will the ladder be trapped in the garage? What is the scenarios from both the garage viewpoint and the ladder viewpoint then? Note: you do not need to answer the questions in the problem (g) in your homework. They are just for you to think of in your spare time.

2. Charge Accelerated in Uniform Electric Field

A charge q will experience a force $F = Eq$ in a uniform electric field of the strength E . This law still holds true in relativistic cases (otherwise you can use it to determine the absolute velocity of a charge). Now you have learnt from the lecture, that the Newtonian law $F = d(mv)/dt$ is modified in relativity that the mass is no longer a constant,

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

Let m_0 be the rest mass of the charge.

(a) Write down the equation of motion for this charge in the relativistic case.

(b) The charge is released at rest when $t = 0$. Determine the velocity of the charge as a function of time by solving the equation of motion. You are encouraged to plot the curve of $v(t)$ out.

(c) If you forget about relativity for the moment, and treat this problem with classical mechanics, then what is the velocity $v_{cl}(t)$ as a function of time.

Note: In classical mechanics, a particle driven by a constant force will accelerate with constant acceleration. Thus after a while, the speed of the particle will exceed the speed of light. However in relativity this is not going to happen, since the particle will become more and more massive as its speed approaching to the speed of light, therefore it becomes harder and harder to keep the acceleration. In this problem, you will find that as $t \rightarrow \infty$, the velocity tends to (but will never reach) the speed of light. This explains that why you cannot push a massive object by constant force to the speed of light. On the other hand, you will see that at the beginning when the velocity is small, the classical solution $v_{cl}(t)$ is a good approximate of the relativistic one $v(t)$. This is why classical mechanics works well in our everyday life.

Chap. 16 Relativistic Energy and Momentum

1. Twin Paradox

You have learnt the twin paradox in the Lecture. Now let us look at an specific example and see how the paradox is solved.

A pair of twins Peter and Paul are born on the day when the spaceship Axiom is leaving for Gliese 876, which is a star 15 light years away from the Earth. Paul boards the Axiom and starts his journey, while Peter stays on the Earth. The spaceship Axiom travels at constant speed $0.6c$, (it is assumed to immediately attain its full speed upon departure), and return

with the same speed as soon as it arrives at Gliese 876.

- (a) How many years it takes for the Axiom to finish a round trip from the frame of the Earth?
- (b) How far is Gliese 876 away from the Earth from the frame of the Axiom? Then how many years it takes for a round trip from the frame of the Axiom?
- (c) When the Axiom returns, how old is Peter (the one staying on the Earth), and how old is Paul (the one travelling with the Axiom)?

As Peter grows up, he shows extraordinary talent in physics. He enters California Institute of Technology at the age of 16 (16 years after his birth) for his undergraduate study. He becomes a full professor at Princeton University when he is 34. Peter has made great achievements in physics, and is finally awarded the Nobel Prize when he is 50.

- (d) Write down the space-time coordinates of the following events in the frame of the Earth, and mark them on the space-time diagram. (The event A is chosen as the origin).

- (1) the event A: Peter and Paul are born,
- (2) the event B: Peter enters California Institute of Technology,
- (3) the event C: Peter becomes a full professor at Princeton,
- (4) the event D: Peter is awarded the Nobel Prize,
- (5) the event E: Paul arrive at the star Gliese 876.

Also draw the world line of Peter and Paul respectively on the same space-time diagram.

- (e) For the events A, B and E, use Lorentz transformation to transform their space time coordinates from the frame of the Earth to that of the outward Axiom.

For the events C and D, we should no longer use the ordinary Lorentz transform to switch to Paul's frame. Because Paul is using two inertial frames: one on the way towards Gliese 876 and the other on the way back to the Earth. When the Axiom turns around at Gliese 876, Paul is switching from one frame to another, and thus his clock should be calibrated again by events D and E.

- (f) Write down the space-time coordinates of events D and E in the frame of the returning Axiom.

The Lorentz transform from the Earth frame to the returning Axiom frame has a shift in time due to the re-calibration of clock. It takes the following form,

$$\begin{cases} x' = \gamma(x + \beta(t - t_1)) \\ t' - t_2 = \gamma((t - t_1) + \beta x) \end{cases}$$

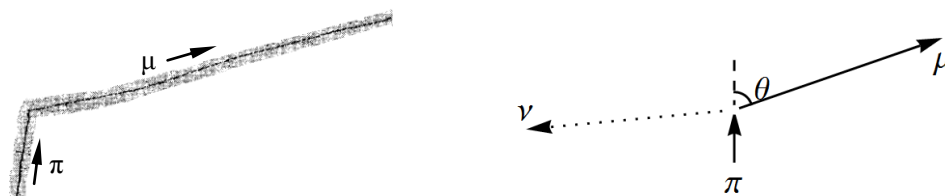
- (g) Use the space-time coordinates of events D and E you have obtained from both frames to determine the shifts t_1 and t_2 . Use this shifted Lorentz transformation (known as Poincaré transform) to transform the event C into the frame of returning Axiom.

- (h) Mark the events A, B, C, D and E on the space-time diagram in Paul's frame. Then draw the world line of Peter and Paul respectively on the same space-time diagram. Try to explain the scenario from Paul's point of view.

Hint: Einstein, Born and Møller invoked gravitational time dilation to solve the twin paradox. The asymmetry between Peter and Paul lies in the effect of acceleration when the Axiom turns around. In general relativity, such an accelerating frame is equivalent to a inertial frame in a gravitational field. The clock farther away from the source of gravity tends to go faster. This effect becomes so remarkable at large distance that it is even possible for the time on the Earth to elapses 18 years within several minutes on the Axiom. This explains why events B and C are simultaneous in Paul's frame.

2. Determine the Rest Mass of π Meson

The pion (π meson) was first discovered on photographic plates in which there were tracks of muons (μ): some unknown particle had come in and decayed, and where its track ended, there was a new track coming off whose properties were found to be those of a muon. It was presumed that a neutrino went off in the opposite direction (leaving no track, because it is neutral). A typical decay process: $\pi^+ \rightarrow \mu^+ + \nu_\mu$. The tracks are shown in the following figure.



The rest energy of the muon was known to be 105.7 MeV, and its kinetic energy was found from the properties of the track to be 4.5 MeV. The neutrino has almost zero rest mass (< 0.19 MeV).

(a) Suppose that the pion decays at rest, how can we find the rest mass of the pion? Hints: first find the momentum of the muon and the neutrino.

(b) If the pion decays when it is still moving (with some momentum we do not know), we will not be able to determine its rest mass now. But from the photograph we can measure the angle between the muon track and the pion track, which gives $\theta = 71^\circ$. With this experimental fact, can you give a lower bound of the pion rest mass (i.e. what is the rest mass at least)?

Chap. 17 Space-Time

1. Four-vector Form of Relativistic Mechanics

Many confusions in our study of relativity arise from the inappropriate human language. In the Lecture, we have learnt that both the length contraction and the time dilation are "apparent" effects, if we perceive Lorentz transformation as a kind of rotation in space-time, then each length and each time interval we measured individually are just apparent length and apparent time, which are not fundamental properties of the object. Therefore in relativity, we must always keep in mind that space and time is a unity. To carry through this idea, we insist the 4-vector notation in any possible cases, such as the space-time position $x_\mu = (x, y, z, t)$ and momentum-energy $p_\mu = (p_x, p_y, p_z, E)$.

But when we come to the concept of velocity, a 3-vector defined as the time derivative of the space, we feel uncomfortable with the unequal treatment of space and time (why time should be the one to differentiate). It is necessary to extend velocity to a 4-vector form, known as the proper velocity, defined as the space-time x_μ differentiated by the proper time τ ,

$$v_\mu = \frac{dx_\mu}{d\tau} = \left(\frac{dx}{d\tau}, \frac{dy}{d\tau}, \frac{dz}{d\tau}, \frac{dt}{d\tau} \right).$$

Here the original time derivative is replaced by that of proper time, which is the invariant interval between space time points,

$$d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2.$$

(a) With the above definitions, write down the components of proper velocity for an object moving along x direction at the (traditional) speed v .

Note: it will be interesting to think of the physical meaning of the time component of the proper velocity. Is it the speed that an object "moving" in the time direction? If you look at the world line of a rest object, you will find that the rest object is indeed moving along the time direction from past to future. This is the famous saying that "the existence itself is already a kind of motion".

Along this line of thought, we can invent the proper acceleration $a_\mu = dv_\mu/d\tau$ and the proper force $F_\mu = dp_\mu/d\tau$, both are the 4-vector generalizations of their 3-vector forms. Then with m being the rest mass, our familiar old friend: Newton's Second Law $F_\mu = ma_\mu$ never breaks down in relativity. Hence we see that there is no need to introduce the "relativistic mass" if we define physical quantities "properly".

To give you a concrete example, let us consider a charge q of rest mass m moving in an uniform magnetic field of the strength B directed along z axis. How much proper force the magnetic field is to exert on the charge? In the classical electromagnetism, we have learnt the formula of Lorentz force $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$, or in its component form

$$\begin{cases} F_x = qv_y B \\ F_y = -qv_x B \end{cases}$$

These two equations still holds in relativity when both F_μ and v_μ are the components of the proper 4-vectors, (with two additional ones $F_z = F_t = 0$).

(b) Write down the equation of motion for the charge in the magnetic field.

(c) With the initial condition given in the problem (a), find the proper velocity $v_\mu(\tau)$ as a function of proper time τ by solving the differential equations.

(d) Integrate $v_\mu(\tau)$ with respect to τ , you will obtain the equation for the world line of the charge. The charge will undergo circular motion. Determine the radius R of the circular orbit and the period T (in time not in proper time) for one rotation.

(e) Compare with the classical cyclotron radius and period you have learnt in the high school, what are the difference? Does this mean that the charge becomes more massive with higher speed?

Note: The concept of relativistic mass has caused a lot of controversies in the early years of relativity. Until today, it is still the most intensively attacked concept by the folk physicists. This is why the contemporary physicists like Taylor and Wheeler^[1] avoid using the this concept altogether: "The concept of "relativistic mass" is subject to misunderstanding. That's why we don't use it. First, it applies the name mass – belonging to the magnitude of a 4-vector – to a very different concept, the time component of a 4-vector. Second, it makes increase of energy of an object with velocity or momentum appear to be connected with some change in internal structure of the object. In reality, the increase of energy with velocity originates not in the object but in the geometric properties of space-time itself."

[1] Taylor, E. F., Wheeler, J. A. (1992), *Spacetime Physics, second edition*